Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Fruit Development

- **Fruit development stages:** Flowering and fruit set, fruit initiation, cell division, fruit enlargement, maturation, ripening, and attainment of harvest maturity constitute the documented developmental sequence in tomato (*Solanum lycopersicum*). Experimental evaluations of dual-purpose hybrids recorded fruit retention intact on plants up to 110 days after transplanting in the main field, facilitating a once-over harvest for industrial processing. Total crop duration from sowing to maturity is documented as 180–190 days in specific cultivars. Stage durations for individual cell division, cell expansion, and fruit enlargement phases are Not specified in the provided sources.

- **Fruit set and initial development:** Pollination, fertilization, flower retention, and initial fruit development are heavily influenced by environmental temperatures and plant growth regulators. Application of parachlorophenoxyacetic acid (PCPA) as a foliar spray at 50–100 mg/litre during flowering promotes fruit set under extreme low and high temperatures. Ethephon applied as a foliar spray at 100–500 mg/litre influences flowering, fruiting, and overall yield. Gibberellic acid (GA) foliar spray at 10–20 mg/litre enhances fruit yield under low temperatures. Documented flower retention rates and exact fruit set percentages under standard conditions are Not specified in the provided sources.

- **Environmental requirements:** Tomato requires a warm climate with an optimal temperature range of
(25–27°C
)$and well-drained soils rich in organic matter. Ambient temperatures exceeding$
(40°C
)$significantly increase flower and fruit drop. Low or high temperatures during fruit set disrupt polli$

- **Water and irrigation management:** Tomato is highly sensitive to soil moisture dynamics. Light irrigation is provided 3–4 days after transplanting, followed by regular irrigation intervals of 7–8 days during the kharif season, 10–12 days during the rabi season, and 5–6 days during the summer season. Flowering and fruit development are critical growth stages where water stress must be strictly avoided. Heavy watering following a prolonged dry spell causes severe fruit cracking, whereas waterlogging increases catfacing deformities and root damage. Drip irrigation minimizes leaf wetness and maintains soil aeration. During extreme heatwave conditions exceeding
(40°C

$), providingsufficientirrigationisrequiredtomitigatethermalstressandpreventflowerandfruita$

- **Nutrient management:** Soil fertility status dictates nutrient application schedules. Standard tomato varieties require 40:24:24 kg N:P:K/acre (100:60:60 kg/ha), whereas hybrid cultivars require 60:36:36 kg N:P:K/acre (150:90:90 kg/ha). Organic enrichment involves incorporating well-rotten farmyard manure (FYM) at 8–10 t/acre (20–25 t/ha) 2–3 weeks before transplanting or vermicompost at 5 t/acre 1 week before transplanting. For varieties, 50% N is applied as a basal dose, followed by top-dressing remaining P, K, and 60 kg N/ha at 30 days after transplanting, and a third N dose of 13.5 kg N/acre (60 kg N/ha) at 50–60 days after transplanting. For hybrids, the second N dose (20 kg N/acre) and potassic fertilizers are top-dressed 30 days after transplanting, and the third N dose (20 kg N/acre) is applied 60 days after transplanting. Soluble fertigation schedules supply split doses every 3 days from flowering set through fruit picking. Chlorosis is corrected by foliar sprays of 1%
($FeSO_4$
$) + 0.5\%$
($ZnSO_4$
$)at60and90daysafterplanting.Localizedcalciumdeficiencycausesblossom-endrot, excessnitrog$
$endrot, graywall, andpuffiness, whileadequatepotassicfertilizationreducesgraywallandinterna$

- **Fruit growth and enlargement:** Measured physical traits across cultivar evaluations document fruit length ranging from 5.17–7.17 cm, fruit width from 5.17–7.67 cm, pericarp thickness from 0.50–0.97 cm (0.63–0.90 cm in processing hybrids), locule count from 2.00–6.00, and fruit density from 7.56–13.87 fruits/kg. Average fruit weight ranges from 72.12–132.28 g in measured hybrid trials, with a target weight of 70–90 g for dual-purpose hybrids. Mean crop productivity in national statistics is reported at 25 t/ha, whereas high-yielding F1 hybrids record yield potentials of 80–90 t/ha (up to 93.2–112.8 t/ha in experimental trials). Fruit volume and daily growth rates are Not specified in the provided sources.

- **Fruit physiological processes:** Physiological changes during fruit enlargement and maturation include dry matter accumulation, cell expansion, sugar and acid accumulation, and lycopene pigment synthesis. In dual-purpose processing hybrids, fine pulp lycopene content ranges from
(>12–13.41 mg/100 g
) fresh weight (
(>14 mg/100 g
) in paste), total soluble solids (TSS) range from 4.5–5.2
(°Brix
$)infreshfruit(8.80 - -11.20$
(°Brix
$)inpureeand$
$(> 26 - -27°$Brix
$)inpaste), titratableacidityrangesfrom0.32--0.83\%(ascitricacid), andpHrangesfrom4.0-$
$-4.3.Specificbiochemicalenzymaticpathwaysduringcellexpansionare Notspecifiedintheprovide$

- **Pollination and fruit retention management:** Exogenous plant growth regulators and cultural practices manage pollination, fruit set, and retention under environmental stress. PCPA foliar spray at 50–100 mg/litre at flowering promotes fruit set under low and high temperatures. Gibberellic acid (GA) foliar spray at 10–20 mg/litre increases yield under cool conditions. Ethephon foliar spray at 100–500 mg/litre improves flowering, fruiting, and yield. Maintaining adequate soil moisture during high-temperature periods (
($>40°C$
$))prevents floral abortion and fruit drop. Inadequate pollination or fertilization results in seed set fa$

- **Canopy and plant management:** Staking tomato plants improves fruit quality by keeping fruits and foliage off the ground, reducing leaf wetness, enabling adequate air circulation, and ensuring uniform spray coverage. Canopy operations include tying, training, periodic clipping of lower leaves, and applying organic mulch around plants to protect root systems and prevent soil-borne fungal spores from splashing onto developing fruits.

- **Pest and disease management:** Major insect pests damaging developing fruits include tomato fruit borer / gram pod borer (*Helicoverpa armigera*), causing 85–93.7% yield loss by boring into fruits; South American tomato pinworm (*Phthorimaea / Tuta absoluta*), causing 60.08–82.31% loss by mining green fruits and creating pinhole entries for secondary fungal rot; tobacco caterpillar (*Spodoptera litura*), boring irregular holes into fruits; and silverleaf whitefly (*Bemisia tabaci*), causing irregular ripening. Integrated management includes field sanitation; raising African marigold 15 days prior as an ovipositional trap crop for *Helicoverpa*; intercropping or guard rows with maize, sorghum, cowpea, or castor; installing yellow/blue sticky traps @ 10/ha (or 4–5/acre), fruit fly traps @ 20–25/ha, and pheromone traps @ 4–5/acre; erecting bird perches @ 20/acre; releasing egg parasitoids *Trichogramma pretiosum* @ 20,000–40,000/acre/week (or 50,000/ha) 4–6 times starting at flower initiation; applying neem cake @ 100 kg/acre, NSKE 5%, *Bacillus thuringiensis* var. *kurstaki* @ 500 g/ha, or label-claimed insecticides. Fruit diseases include early blight (*Alternaria solani*), late blight, Fusarium wilt, Sclerotium rot, bacterial wilt, and tomato leaf curl virus, managed by planting triple-disease-resistant hybrids (Arka Apeksha, Arka Vishesh, Arka Rakshak, Arka Samrat), seed treatment with Captan 75% WS @ 20–30 g/kg seed or *Trichoderma viride* 1% WP @ 9 g/kg seed, and sprays of copper oxychloride, mancozeb, or azoxystrobin.

- **Physiological disorders:** Documented disorders affecting developing fruits include blossom-end rot (BER), caused by localized calcium deficiency and exacerbated by low soil Ca, excessive N, ammoniacal N sources, high soluble K/Mg, high salinity, low humidity, moisture fluctuations, and root damage, managed by Ca soil application and good water management; fruit cracking (radial and concentric), caused by rapid internal tissue expansion exceeding epidermal expansion due to moisture fluctuations and sun exposure, managed by tolerant cultivars, good canopy cover, and stable irrigation; sunscald, caused by direct fruit exposure to intense sunlight, managed by maintaining foliage cover; catfacing, characterized by gross blossom-end deformity during flower formation due to low temperatures, herbicide drift, or thrips feeding, managed by tolerant varieties and avoiding wa-

terlogging; puffiness, featuring internal cavities between seed gel and outer wall due to pollination failure or extreme temperatures, managed by balanced N fertilization; gray wall (blotchy ripening), showing vascular tissue necrosis and grayish appearance under cool, short days and high N, managed by adequate potassic fertilization; irregular ripening, exhibiting star-burst color patterns caused by whitefly nymph feeding, managed by whitefly control; internal white tissue, forming hard white tissue under high ripening temperatures, managed by potassic fertilization; pox and fleck, showing cuticular disruptions and green/gold spots due to genetic-environment interactions, managed by tolerant varieties; zebra stripe, forming dark green spots/lines from stem to bloom end, managed by variety selection; and zippering, producing longitudinal scars from anther adhesion, managed by variety selection.

- **Fruit quality attributes:** Documented quality characteristics for fresh market and industrial processing include average fruit weight (70–90 g target), fruit firmness (4.0–9.0 kg/
(cm^2
), $withatarget$
($\geq$ 8 kg/cm^2
)$fordual-purposehybridsand$
($>$ 4 kg/cm^2
)$forindustrialprocessing$), $totalsolublesolids$(4.5 $-$ $-$5.2
(°Brix
)$infreshfruit$, 8.80 $-$ $-$11.20
(°Brix
)$inpuree$,
($>$ 26 $-$ $-$27°Brix
)$inpaste$, $withanindustrialminimumtargetof$
(4.2°Brix
)), $titratableacidity$(0.32 $-$ $-$0.83%$ascitricacid$, $withanindustrialtarget$
($\leq$ 0.40%
)), pH(4.0 $-$ $-$4.4, $withanindustrialrequirement$
($<$ 4.5
)), $Huntercolorvalue$
($a/b \geq 2.0$
)(1.96 $-$ $-$2.12$inpuree$), $lycopenecontent$(
($>$ 12 $-$ $-$13.41 mg/100 g
)$freshweightinfinepulp$,
($>$ 14 mg/100 g
)$inpaste$), $andpresenceofjointlesspedicel$(
(j_2
))$forstemlessfruitdetachment$.

- **Fruit maturation and ripening:** Ripening stages progress from immature green to mature green, breaker stage (color change initiating at blossom end), and vine-ripe. Pre-harvest foliar spray of ethephon at 1,000 mg/litre accelerates fruit ripening. Commercial mature green fruits are ripened artificially and uniformly by dipping in ethephon 39% SL @ 2500 ppm. Mature green fruits can be stored for up to 30 days at

(10°C
), $whereasfullyripefruitsretainqualityforapproximately7days(1week)$.

- **Harvest maturity indicators:** Harvest readiness criteria depend on market destination. Fresh market tomatoes picked at the mature green stage are harvested when approximately 10% of the fruits on the first hand reach the breaker stage of maturity. Vine-ripe tomatoes are hand-harvested every other day as color appears. Immature green fruits should not be harvested because low sugar and acid levels result in poor flavor even after ethylene gassing; mature green fruits harvested when dry develop full flavor comparable to vine-ripened fruit within 24 hours of color appearance. Picking frequency is every 2–3 days in warm weather and weekly in cool weather, executing removal by twisting the fruit rather than pulling.

- **Monitoring and scheduling:** Crop operational schedules mandate irrigation every 5–6 days in summer, 7–8 days in kharif, and 10–12 days in rabi; fertigation split applications every 3 days; pheromone lure replacement every 2–3 weeks; field scouting for AESA and Pest-to-Defender ratios; weekly release of *Trichogramma* parasitoids for 4–6 weeks starting at flower initiation; and harvesting at 2–3 day intervals in warm weather or weekly in cool weather.

- **Effects on crop performance:** Severe abiotic stress (temperatures
(>40°C
)$ordrought)ornutrientimbalances(highN, deficientCaorK)triggerflowerabortion, fruitdrop, a$
$-93.7\%loss)orTuta\ absoluta(upto60.08--82.31\%loss)drasticallylowerstotalyieldandmarketabil$
$disease-resistanthybridscombinedwithintegratednutrient, irrigation, andpestmanagementstab$
$-90t/haandpreservesprocessingqualityattributes.$

- **Integrated workflow:** The documented operational sequence spans: (1) presowing deep summer ploughing, field sanitation, and soil solarization; (2) nursery bed preparation and seed treatment with Captan or *Trichoderma viride*; (3) field preparation with FYM/vermicompost incorporation and basal fertilizer application; (4) transplanting onto raised beds with light establishment irrigation; (5) vegetative canopy management through staking, tying, lower leaf clipping, organic mulching, and fertigation; (6) flowering and fruit set management using growth regulators (PCPA, *GA*, ethephon), drip irrigation, marigold trap cropping, sticky/pheromone traps, and *Trichogramma* releases; (7) fruit enlargement under high potassic fertigation and disorder monitoring; (8) harvesting mature green fruits at 10% first-hand breaker stage by twisting dry fruits at 2–3 day intervals; and (9) post-harvest uniform ripening via ethephon 39% SL @ 2500 ppm dip or cool storage at
(10°C
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